

AUDISANKARA COLLEGE OF ENGINEERING AND TECHNOLOGY

Gudur, Andhra Pradesh

Department of Artificial Intelligence

PROJECT DOCUMENTATION

Credit Card Approval Prediction using Machine Learning

Field	Details
Student Name	Mulammagari Manoj Kumar Reddy
Mode	Individual Project (Self-guided)
Department	B.Tech, Artificial Intelligence
Domain	Artificial Intelligence and Machine Learning
Internship Program	SmartBridge / SkillWallet AI-ML Virtual Internship
Live Application	credit-card-approval-prediction-2-mfxo.onrender.com
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1. Introduction

1.1 Project Description

Credit Card Approval Prediction is a machine learning system that automates the decision-making process behind approving or rejecting credit card applications. Financial institutions traditionally assess applicant details such as income, employment status, existing debt, and credit history manually — a process that is slow, inconsistent, and prone to bias. This project builds a supervised ML pipeline that learns patterns from historical applicant data (`clean_dataset.csv`) and predicts whether a new application is likely to be Approved or Rejected, covering data preprocessing, feature engineering, model training and evaluation, and deployment as a live web application.

1.2 Problem Statement

Manual credit card approval processes are inconsistent, slow, and prone to subjective bias, which can result in creditworthy applicants being rejected or high-risk applicants being approved. A data-driven, automated classification system improves decision speed, consistency, and reliability.

1.3 Objectives

- Analyze and preprocess a real-world credit card application dataset (`clean_dataset.csv`) for machine learning.
- Engineer meaningful features, such as the Debt-to-Income Ratio, to improve predictive performance.
- Train and compare Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers.
- Evaluate models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC.
- Deploy the best-performing model as an accessible web application for real-time predictions.

1.4 Scope

The scope is limited to binary classification of credit card applications (Approved / Rejected) using the provided dataset. The system is an academic, end-to-end ML workflow demonstration — from preprocessing to deployment — and is not intended for production use in an actual financial institution without further regulatory, fairness, and security review.

2. Technical Architecture & Workflow

2.1 Architecture Flow

```
Raw Dataset (clean_dataset.csv)
    |
    v
Data Preprocessing (missing/duplicate check, one-hot
encoding, log transform, feature engineering, scaling)
    |
    v
Train-Test Split (80/20)
    |
    v
Model Training (Logistic Regression | Decision Tree |
                Random Forest | XGBoost)
    |
    v
Model Evaluation (Accuracy, Precision, Recall,
                 F1-Score, Confusion Matrix, ROC-AUC)
    |
    v
Best Model Selection --> Web Application (Flask)
```

2.2 Pre-requisites

- Python 3.x, Pandas, NumPy
- scikit-learn, XGBoost
- Matplotlib / Seaborn for EDA
- Flask for the prediction web service
- Git for version control

2.3 Project Workflow (Milestones)

Milestone	Summary
1. Data Collection & Understanding	Loaded clean_dataset.csv (690 records, 16 columns); ran EDA on feature distributions and class balance; identified categorical/numerical features.
2. Preprocessing & Feature Engineering	Handled missing values & duplicates; encoded categorical fields; log-transformed Income; engineered Debt-to-Income Ratio; 80/20 split; applied StandardScaler.
3. Model Building	Trained Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers.
4. Evaluation & Selection	Compared models on Accuracy, Precision, Recall, F1, Confusion Matrix, ROC-AUC; selected XGBoost (89.86% accuracy).
5. Deployment	Saved model with joblib; built Flask backend + HTML/CSS/JS frontend; deployed on Render; tested end-to-end.

3. Methodology

3.1 Dataset Description

Attribute	Details
Dataset Name	clean_dataset.csv
Records / Columns	690 records, 16 columns
Type	Structured / Tabular data
Target Variable	Credit Card Approval Status (Approved / Rejected)
Feature Types	Categorical and numerical applicant attributes

3.2 Data Preprocessing Steps

- Loaded the dataset with Pandas and inspected structure using `df.head()`, `df.info()`, `df.describe()`.
- Verified missing values and checked/removed duplicate records to prevent data leakage.
- Encoded categorical features (Industry, Ethnicity, Citizen) using Label/One-Hot Encoding.
- Applied a log transformation to Income to reduce right-skewness from high-income outliers.
- Engineered a Debt-to-Income Ratio feature to better capture applicant financial risk.
- Split the data into training (80%) and testing (20%) sets, then applied StandardScaler where required (e.g. Logistic Regression).

3.3 Algorithms Used

Algorithm	Purpose / Rationale
Logistic Regression	Baseline linear classifier; interpretable and fast.
Decision Tree	Captures non-linear relationships and feature interactions.
Random Forest	Ensemble of trees; reduces overfitting, improves robustness.
XGBoost	Gradient boosting; typically strongest on structured/tabular data.

3.4 Evaluation Metrics

- Accuracy — overall proportion of correctly classified applications.
- Precision & Recall — correctness of predicted approvals vs. actual approvals identified.
- F1-Score — harmonic mean of Precision and Recall.
- Confusion Matrix — breakdown of true/false positives and negatives.
- ROC-AUC — ability to distinguish Approved vs. Rejected across thresholds.

4. Results

4.1 Model Comparison

All four classification algorithms were trained on the preprocessed dataset and evaluated on the held-out test set:

Model	Accuracy
Logistic Regression	87.68%
Decision Tree	84.78%
Random Forest	88.41%
XGBoost (Best Model)	89.86%

The XGBoost Classifier achieved the highest test accuracy of 89.86% and was selected as the final model for deployment. Its gradient boosting approach captured non-linear relationships between applicant features (Income, Debt, Credit Score, Prior Default) more effectively than the other three algorithms.

4.2 Best Model — Detailed Evaluation (XGBoost)

Metric	Value
Accuracy	89.86%
Precision (weighted avg)	0.90
Recall (weighted avg)	0.90
F1-Score (weighted avg)	0.90
ROC-AUC Score	0.9453

Confusion Matrix (XGBoost)

	Predicted: Rejected	Predicted: Approved
Actual: Rejected	71 (True Negative)	6 (False Positive)
Actual: Approved	8 (False Negative)	53 (True Positive)

4.3 Exploratory Data Analysis

EDA was performed on `clean_dataset.csv` (690 records, 16 features) using Pandas, Matplotlib, and Seaborn — examining structure with `df.info()/df.describe()`, the distribution of Approved vs. Rejected applications, relationships between Income, Debt, Age and approval outcomes, and a correlation heatmap. This directly informed the log transformation of Income and the Debt-to-Income Ratio feature.

5. Web Application

The trained XGBoost model has been deployed as an interactive web application, “Credit Card Approval Prediction”, built with Flask (backend) and HTML/CSS/JavaScript (frontend). Users enter applicant details through a form and receive an instant Approved / Not Approved prediction along with a confidence score.

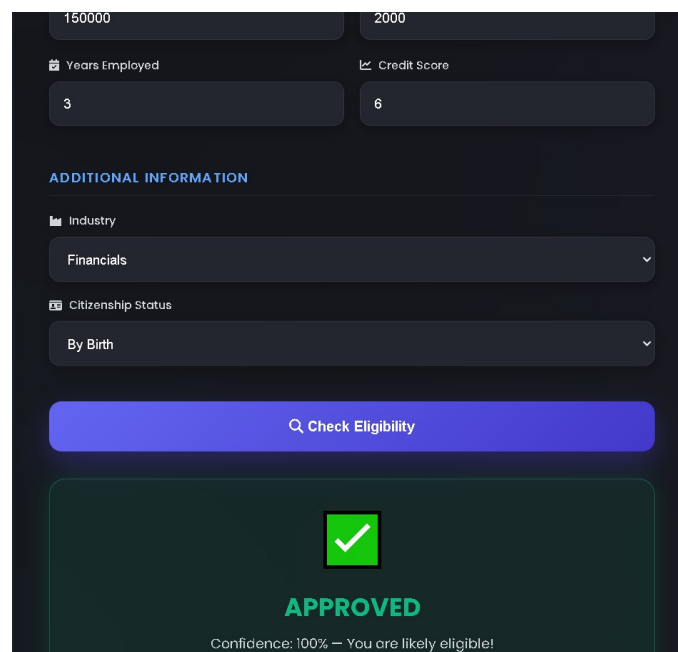
Live at: <https://credit-card-approval-prediction-2-mfxo.onrender.com>

5.1 Application Form

The form collects: Gender, Age, Marital Status, Bank Customer status, Employment status, Prior Default status, Annual Income, Total Debt, Years Employed, Credit Score, Industry, and Citizenship Status.

5.2 Prediction Result

On submission, form data is sent via a POST request to the /predict endpoint. The Flask backend encodes categorical fields, builds the feature vector in the order expected by the model, and returns a prediction with a confidence score, shown instantly with a colour-coded status: green for Approved, red for Not Approved.



The screenshot displays the web application interface. At the top, there are input fields for 'Annual Income' (150000) and 'Total Debt' (2000). Below these are 'Years Employed' (3) and 'Credit Score' (6). A section titled 'ADDITIONAL INFORMATION' contains a dropdown for 'Industry' (Financials) and a dropdown for 'Citizenship Status' (By Birth). A blue button labeled 'Check Eligibility' is positioned below the form. The result area at the bottom features a large green checkmark icon, the word 'APPROVED' in green, and the text 'Confidence: 100% — You are likely eligible!'.

Figure 1: Prediction result for an applicant with a strong financial profile — classified as Approved with 100% confidence.

Figure 2: Prediction result for an applicant with a weaker financial profile — classified as Not Approved (Confidence: 79.2%).

5.3 Backend Implementation (app.py)

The Flask backend loads the trained XGBoost model, its label encoders, and feature column order, then exposes / (the form) and /predict (JSON in, prediction out):

```
model = joblib.load('models/card_model.joblib')
encoders = joblib.load('models/encoders.joblib')
feature_cols = joblib.load('models/feature_cols.joblib')

@app.route('/', methods=['POST'])
def predict():
    data = request.get_json()
    input_df = pd.DataFrame([{'Gender': safe_encode('Gender', data['gender']),
                              'Age': float(data['age']), 'Debt': float(data['debt']),
                              'Income': float(data['income']), ...}])
    prediction = model.predict(input_df)[0]
    probability = model.predict_proba(input_df)[0]
    confidence = round(float(max(probability)) * 100, 1)
    return jsonify({'approved': bool(prediction == 1),
                    'confidence': confidence})
```

5.4 Frontend Implementation

The frontend is a single-page HTML interface with custom CSS (dark theme, blue gradient accents) and vanilla JavaScript that submits the form asynchronously via fetch() and renders the Approved/Not Approved result with its confidence score, without a page reload.

6. Conclusion and Future Scope

6.1 Conclusion

This project demonstrates an end-to-end machine learning workflow for predicting credit card approval outcomes. Through systematic preprocessing — missing value checks, duplicate handling, categorical encoding, log transformation of Income, and Debt-to-Income Ratio engineering — the dataset was prepared for effective model training. Four classifiers were trained and evaluated, with XGBoost achieving the highest accuracy (89.86%) and ROC-AUC (0.9453). The trained model was deployed as a live, publicly accessible web application, allowing real-time credit card approval predictions and demonstrating how ML can support faster, more consistent, data-driven credit decisions.

6.2 Future Scope

- Incorporate a larger and more diverse dataset to improve generalization.
- Apply SMOTE or similar techniques to handle class imbalance, if present.
- Perform hyperparameter tuning (GridSearchCV / RandomizedSearchCV).
- Add explainability tools (SHAP, LIME) to improve transparency of credit decisions.
- Implement user authentication and prediction-history tracking.
- Deploy on a scalable cloud platform for production-level use.

7. References

- Scikit-learn Documentation — <https://scikit-learn.org/stable/documentation.html>
- XGBoost Documentation — <https://xgboost.readthedocs.io/>
- Pandas Documentation — <https://pandas.pydata.org/docs/>
- Python Documentation — <https://docs.python.org/3/>
- Live Application — <https://credit-card-approval-prediction-2-mfxo.onrender.com>
- Dataset: clean_dataset.csv (project dataset)